



EVERYDAY IS A PAYDAY

EXECUTIVE SUMMARY

PayDay — An Integrated e-Wallet for MTN Mobile Money, Orange Money and UBA Bank

Republic of Cameroon | Document version 1.0 | August 2026

The Opportunity

Mobile money dominates day-to-day payments in Cameroon, with MTN Mobile Money and Orange Money serving the large majority of the market, while banks such as UBA continue to hold formal savings, salary, and business accounts. Most Cameroonian users and small businesses juggle two or three of these accounts at once, and moving value between them today means agent visits, successive transfer fees, and disconnected apps and USSD menus.

The Solution

PayDay is a single, neutral e-wallet built around what the design documentation calls “the Triangle Model”: one central wallet with a direct, bidirectional connection to each of the three channels — MTN Mobile Money, Orange Money, and UBA. Every transaction is either a deposit (channel → wallet) or a withdrawal (wallet → channel); the three channels never need to talk to each other directly, because PayDay's wallet is always the intermediary and single source of truth for the user's balance.

Value Proposition

PayDay unifies Cameroon's fragmented mobile-money landscape into a single wallet. Instead of juggling separate MTN Mobile Money, Orange Money, and UBA balances — each requiring agent visits, multiple transfer fees, and disconnected apps — users get one central wallet that deposits from and withdraws to all three channels directly. Money “lives in one place,” with a unified transaction history and simpler reconciliation, especially valuable for small businesses managing cash across providers. The transaction fee PayDay charges replaces, rather than adds to, the agent commissions and hop-by-hop transfer costs users already pay today — trading fragmented friction for one seamless, trustworthy financial hub.

Business Model

PayDay earns a small fee on each deposit and/or withdrawal, mirroring the fee structure MTN and Orange already use for cash-in/cash-out. Costs are driven by four lines: API/integration fees paid to

MTN, Orange, and UBA per transaction; cloud hosting and database infrastructure; SMS/notification delivery; and compliance and licensing costs tied to KYC/AML and electronic money regulation. PayDay does not custody funds directly — wallet balances are backed 1:1 by a pooled trust/float account held at a partner bank, keeping PayDay's own software layer focused purely on the ledger.

Competitive Advantage

Advantage	Why it matters
Channel neutrality	Not owned by MTN or Orange, so it has no incentive to lock users into a single network.
Bridges mobile money and formal banking	Connects UBA (formal banking) with MTN and Orange (mobile money) — a harder problem than a mobile-money-only bridge.
Extensible “adapter” architecture	New channels (Express Union, Afriland First Bank, etc.) are added as plug-ins, not rebuilds.
Built-in auditability	Every balance change traces to one transaction record, supporting regulator and partner-bank trust.
Designed for the real device mix	Degrades to a lightweight/USSD-friendly flow, not just a smartphone-only app.

Version 1 Scope

- A single-currency (XAF) wallet per registered user, with deposit and withdrawal across all three channels.
- KYC-based registration, transaction PIN/OTP security, and an auditable transaction ledger.
- An administrative back-office for monitoring, reconciliation, and reporting.
- Wallet-to-wallet transfers, bill payments, merchant payments, and multi-currency support are deliberately deferred to later phases, keeping v1 simple, testable, and easy to reconcile.

Key Considerations

Because PayDay would hold customer funds, even briefly, it must operate as a licensed Electronic Money Institution or in partnership with a licensed bank or EMI — this regulatory path, along with formal partnership agreements with MTN, Orange, and UBA, is the critical dependency for launch. The competitive position described above is a strong starting point but not yet a proven moat; securing exclusive or early partnerships and demonstrating a clear cost advantage over existing agent fees will determine how defensible it becomes.



EVERYDAY IS A PAYDAY

SYSTEM ANALYSIS AND DESIGN DOCUMENTATION

PayDay

*An Integrated e-Wallet for MTN Mobile Money, Orange Money and UBA Bank
Transactions in Cameroon*

*“The Triangle Model” — a simple, single central wallet
interoperable with three payment channels*

Prepared for: Software / Product Team

Region of deployment: Republic of Cameroon

Document version: 1.0

Date: August 2026

Table of Contents

1. Introduction.....	3
1.1 Background.....	3
1.2 Problem Statement.....	3
1.3 Aim and Objectives.....	3
1.4 Scope and Limitations.....	4
2. System Analysis.....	5
2.1 Methodology.....	5
2.2 Requirement Analysis.....	5
2.3 Feasibility Study.....	7
2.4 Use Case Analysis.....	8
2.5 Data Flow Diagrams.....	9
3. System Design.....	11
3.1 The Triangle Model — Design Concept.....	11
3.2 System Architecture.....	11
3.3 Database Design.....	12
3.4 Process Design — Transaction Sequence Flows.....	13
3.5 Interface / API Design.....	15
4. Security and Compliance Considerations.....	17
5. Recommended Technology Stack.....	18
6. Implementation Roadmap.....	19
7. Conclusion.....	20

1. Introduction

1.1 Background

Mobile money has become the dominant channel for digital financial transactions in Cameroon. MTN Mobile Money (MoMo) and Orange Money together serve the overwhelming majority of the mobile-financial-services market, while traditional banks such as United Bank for Africa (UBA) continue to hold formal savings, salary, and business accounts. In practice, a typical Cameroonian consumer or small business today juggles at least two or three of these accounts separately: a mobile money wallet for daily transactions, a bank account for salary and savings, and often a second mobile money wallet on the rival network because friends, family, or business partners use a different operator.

This fragmentation creates friction: transferring value from a bank account to a mobile money wallet (or between the two mobile money operators) usually requires visiting an agent, paying successive transfer fees, or navigating separate USSD menus and apps that do not talk to each other.

PayDay is proposed as a simple, unifying e-banking application that gives every user one central in-app wallet, and lets that wallet send and receive funds with all three channels — MTN Mobile Money, Orange Money, and UBA — directly. Because the wallet sits at the centre of exactly three external relationships, the system's core money-movement model is described in this document as “the Triangle”: the wallet forms one point that connects outward to MTN, Orange, and UBA, and each of those connections is bidirectional (deposit into the wallet, or withdraw out of the wallet, on any of the three channels).

1.2 Problem Statement

Cameroonian users currently lack a single, neutral application that allows them to consolidate and move money freely between MTN Mobile Money, Orange Money, and a bank account (UBA), without relying on physical agents, manual cash-out/cash-in cycles, or multiple disconnected apps. This slows down transactions, increases transfer costs, and makes it difficult for individuals and small businesses to track their overall balance across providers.

1.3 Aim and Objectives

The aim of this project is to design and document a simple, secure e-wallet system that interoperates with MTN Mobile Money, Orange Money, and UBA, allowing users to store, send, and receive money through a single application.

Specific objectives:

- Provide users with a central digital wallet where money can be stored and managed.
- Allow users to fund (deposit into) the wallet from MTN Mobile Money, Orange Money, or UBA.

- Allow users to withdraw money from the wallet back out to MTN Mobile Money, Orange Money, or UBA.
- Maintain an accurate, auditable ledger of every transaction across all three channels.
- Apply strong authentication (PIN/OTP) and basic KYC to protect user funds.
- Keep the first version (v1) of the system intentionally simple: one wallet, three channels, deposit and withdrawal only — leaving wallet-to-wallet transfers between different users, bill payments, and merchant payments for later phases.

1.4 Scope and Limitations

In scope (Version 1 — the Triangle model)

- A single-currency (XAF) in-app wallet per registered user.
- Deposit (cash-in) from MTN MoMo, Orange Money, or UBA into the wallet.
- Withdrawal (cash-out) from the wallet to MTN MoMo, Orange Money, or UBA.
- User registration with basic KYC (ID document, phone number verification).
- Transaction history, receipts and notifications.
- An administrative back-office for monitoring, reconciliation, and reporting.

Out of scope for Version 1 (future phases)

- Wallet-to-wallet transfers between two different PayDay users.
- Bill payments, airtime purchase, and merchant/QR payments.
- Multi-currency support and cross-border transfers.
- Loans, savings products, or interest-bearing balances.

Keeping the scope to “one wallet, three channels, two directions” (deposit and withdraw) is a deliberate design decision to keep the first release simple, testable, and easy to reconcile before more complex features are added.

2. System Analysis

2.1 Methodology

This documentation follows a structured System Development Life Cycle (SDLC) approach, specifically a lightweight Structured Systems Analysis and Design Method (SSADM)-style process: (1) requirements gathering, (2) feasibility study, (3) process modelling using use case diagrams and data flow diagrams (DFDs), and (4) design of the architecture, database, and process/sequence flows. This approach was chosen because the three external integrations (MTN, Orange, UBA) are well-defined, transaction-based interfaces, which are naturally suited to structured, process-oriented modelling rather than a purely object-oriented approach.

2.2 Requirement Analysis

2.2.1 Functional Requirements

ID	Requirement	Priority
FR-01	The system shall allow a new user to register using their name, phone number, email, and a government ID for KYC.	High
FR-02	The system shall allow a registered user to log in securely (password + OTP or biometric).	High
FR-03	The system shall provide each user with exactly one central wallet, created automatically at registration.	High
FR-04	The system shall allow a user to fund the wallet by initiating a deposit from MTN MoMo.	High
FR-05	The system shall allow a user to fund the wallet by initiating a deposit from Orange Money.	High
FR-06	The system shall allow a user to fund the wallet by initiating a deposit from a UBA account.	High
FR-07	The system shall allow a user to withdraw from the wallet to MTN MoMo, Orange Money, or UBA.	High
FR-08	The system shall require a transaction PIN or OTP confirmation for every deposit and	High

ID	Requirement	Priority
	withdrawal.	
FR-09	The system shall update the wallet balance only after a channel confirms success (via callback/webhook).	High
FR-10	The system shall record every transaction with amount, channel, direction, fee, status, and timestamp.	High
FR-11	The system shall notify the user (SMS and/or push) after every completed or failed transaction.	Medium
FR-12	The system shall allow the user to view a paginated transaction history and download a receipt.	Medium
FR-13	The system shall allow an administrator to view, filter, and reconcile transactions across all three channels.	Medium
FR-14	The system shall automatically reverse (roll back) a wallet debit if a withdrawal to an external channel fails.	High

2.2.2 Non-Functional Requirements

ID	Requirement	Detail
NFR-01	Security	All traffic over TLS 1.2+; sensitive data (PIN, ID numbers) encrypted at rest (AES-256); passwords hashed with bcrypt/argon2.
NFR-02	Availability	Target 99.5% uptime for the core wallet and API gateway; channel outages must degrade gracefully (queue & retry).
NFR-03	Performance	95% of deposit/withdraw requests should receive an initial acknowledgement within 3 seconds; final confirmation depends on the external channel.

ID	Requirement	Detail
NFR-04	Auditability	Every balance change must be traceable to exactly one transaction record; no direct balance edits outside the ledger service.
NFR-05	Compliance	Alignment with BEAC/COBAC electronic money regulations and each partner's (MTN, Orange, UBA) API terms of use and KYC/AML rules.
NFR-06	Usability	Simple, three-step flow (choose channel → enter amount → confirm PIN) usable by first-time smartphone users; app should also degrade to a lightweight/USSD-friendly flow where feasible.
NFR-07	Scalability	Stateless application servers behind a load balancer so capacity can grow horizontally as users increase.

2.3 Feasibility Study

Technical feasibility

MTN Mobile Money exposes a Collections and Disbursements REST API (MoMo API), Orange Money exposes a Web Payment / Orange Money API, and UBA offers corporate/API banking integration (direct API partnership or an aggregator such as a licensed payment service provider). All three support server-to-server callbacks/webhooks, which the proposed architecture relies on to confirm transactions asynchronously. This is technically feasible using standard REST/JSON integration patterns.

Operational and regulatory feasibility

Because PayDay would hold customer funds (even briefly, in the wallet), it must operate either as a licensed Electronic Money Institution (EMI) under COBAC/BEAC regulations, or in partnership with a licensed bank or EMI that holds the pooled float account on the system's behalf. This documentation assumes the latter, safer path for a first version: PayDay's own wallet balances are backed 1:1 by a pooled trust/float account held at a partner bank (which could be UBA itself), while PayDay's software layer only manages the ledger, not the actual custody of funds.

Economic feasibility

Costs are primarily: (a) integration/API fees charged by MTN, Orange and UBA per transaction, (b) cloud hosting and database costs, (c) SMS/notification gateway costs, and (d)

compliance/licensing costs. Revenue can come from a small transaction fee on deposits and/or withdrawals, similar to existing mobile money cash-in/cash-out fee structures.

2.4 Use Case Analysis

The core actors are the Customer (registered wallet user), the System Administrator, and the three external systems (MTN MoMo API, Orange Money API, UBA API), which are treated as actors from the perspective of the wallet system because they are systems outside PayDay's control.

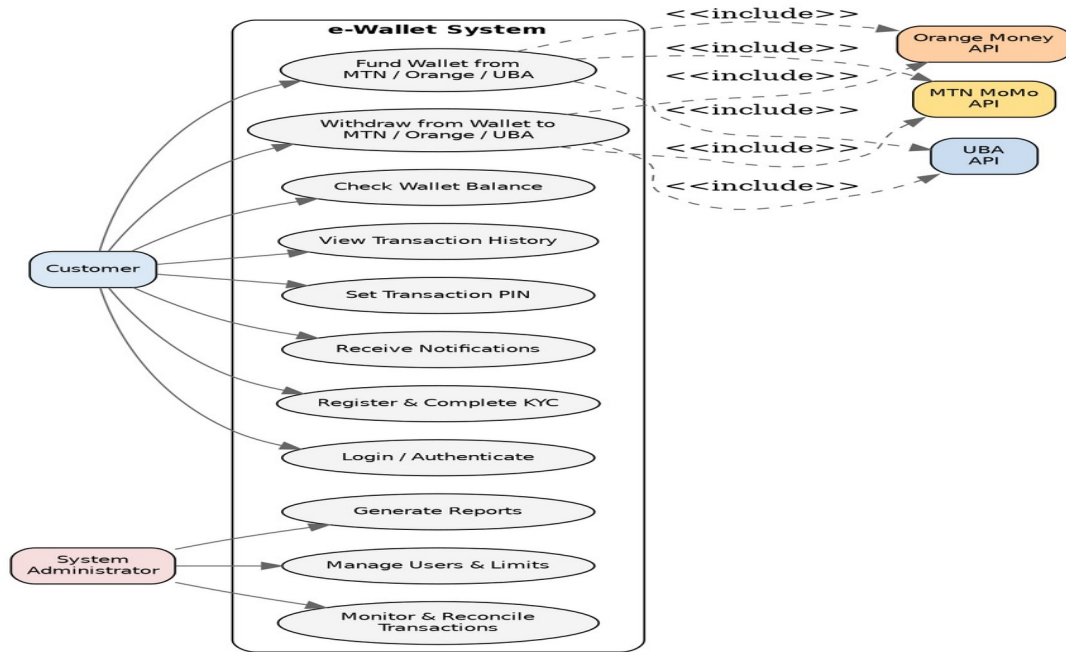


Figure 1: Use Case Diagram — PayDay

2.4.1 Use case descriptions (key cases)

Use Case	Actor(s)	Description	Pre-condition
Fund Wallet	Customer, MTN/Orange/UBA API	User selects a channel and amount; system requests a collection from that channel and credits the wallet once confirmed.	User is logged in and KYC-verified.
Withdraw from Wallet	Customer, MTN/Orange/UBA API	User selects a channel and amount; system debits the wallet (pending) and requests a disbursement to that channel.	Sufficient wallet balance; PIN set.
View Transaction History	Customer	User views a chronological list of deposits and withdrawals	User is logged in.

Use Case	Actor(s)	Description	Pre-condition
		with status.	
Manage Users & Limits	Administrator	Admin views/suspends accounts, sets daily/monthly transaction limits.	Admin authenticated.
Monitor & Reconcile	Administrator	Admin compares PayDay's ledger against each channel's settlement report to detect mismatches.	Settlement files/reports available.

2.5 Data Flow Diagrams

2.5.1 Context Diagram (Level 0)

The context diagram treats the entire PayDay system as a single process and shows how data flows between it and each external entity: the Customer, the Administrator, and the three payment channels.

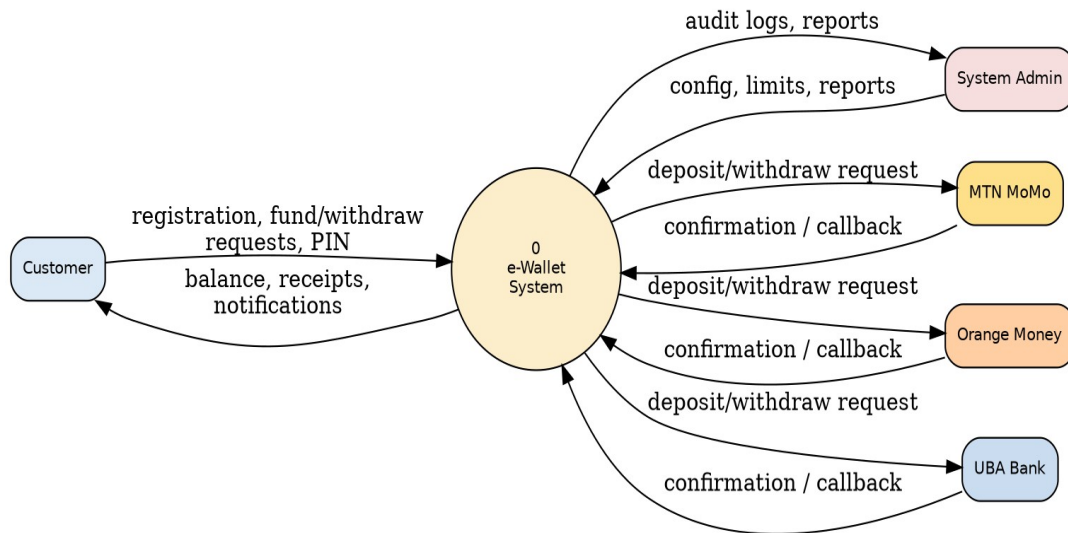


Figure 2: Level-0 (Context) Data Flow Diagram

2.5.2 Level-1 Data Flow Diagram

The level-1 diagram decomposes the system into its four core processes: authentication/KYC, fund/withdraw processing, wallet ledger update, and reconciliation/notification — together with the data stores each process reads or writes.

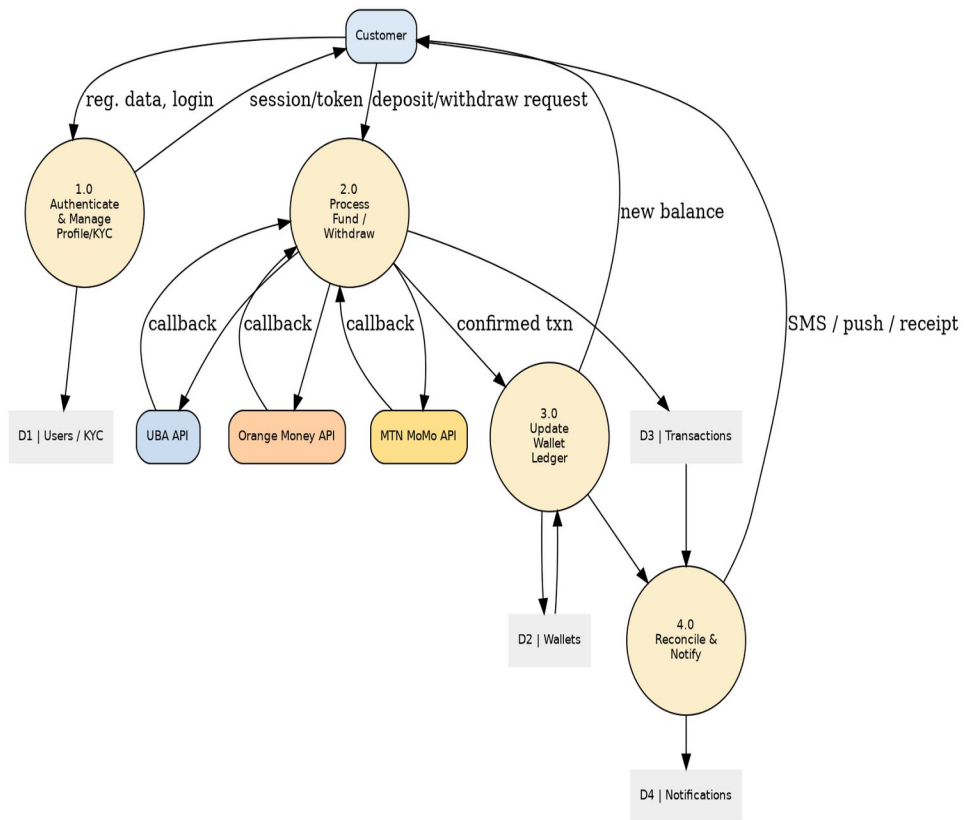


Figure 3: Level-1 Data Flow Diagram

3. System Design

3.1 The Triangle Model — Design Concept

At the heart of PayDay's design is a deliberately simple topology: one central wallet with exactly three bidirectional relationships. This is illustrated below as a triangle, with the wallet as the anchor point and each payment channel as a vertex. Every transaction in the system is either money flowing inward (deposit, channel → wallet) or outward (withdrawal, wallet → channel) along one edge of the triangle. No direct edges exist between the three channels themselves — MTN, Orange and UBA never talk to each other directly; the wallet is always the intermediary and the single source of truth for the user's balance.

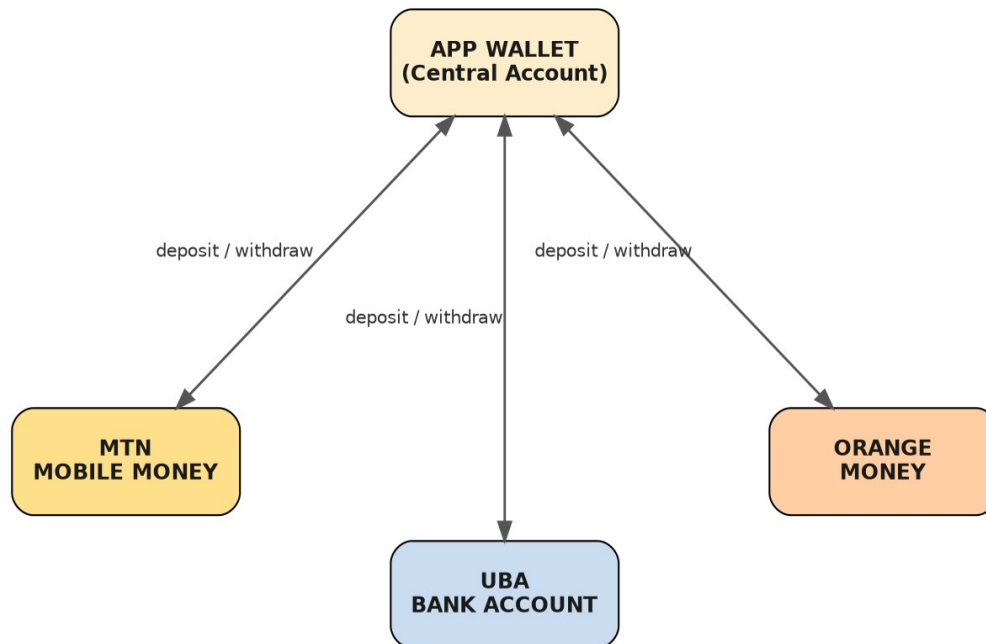


Figure 4: The Triangle Model — Wallet at the centre of three payment channels

This design keeps reconciliation simple (every movement of money is wallet-centred), keeps the mental model simple for users (“my money lives in one place”), and keeps the codebase simple because a single Transaction Manager module can treat MTN, Orange, and UBA as interchangeable “channel adapters” that implement the same two operations: initiate collection (deposit) and initiate disbursement (withdrawal).

3.2 System Architecture

PayDay follows a layered, service-oriented architecture. The mobile/web client talks only to PayDay's own API Gateway; it never talks directly to MTN, Orange, or UBA. All three external integrations are isolated behind channel adapters inside the Transaction Manager, so that adding a fourth channel in future (e.g., Express Union, Afriland First Bank) means writing one new adapter rather than changing the core wallet logic.

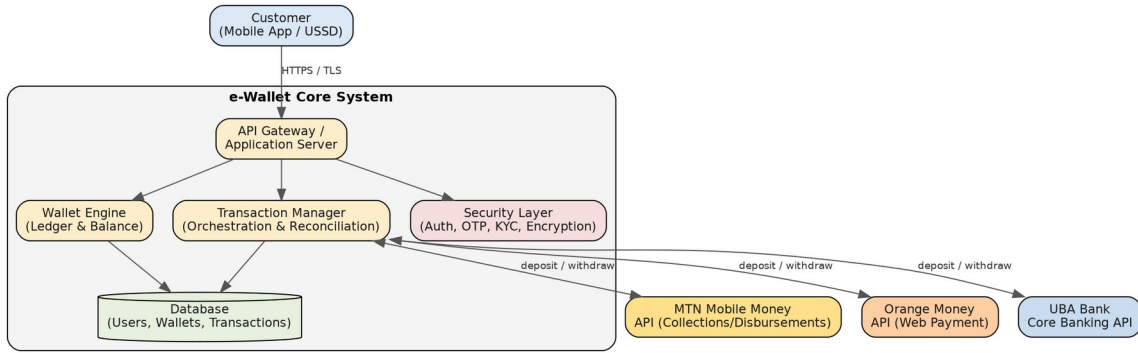


Figure 5: High-Level System Architecture

Architectural layers

Layer	Responsibility
Client (Mobile App / USSD)	User interface for registration, login, fund/withdraw requests, balance & history.
API Gateway	Single entry point; TLS termination, rate limiting, request routing, JWT/session validation.
Security Layer	Authentication, OTP/PIN verification, KYC checks, encryption of sensitive fields, fraud rules.
Wallet Engine	Owns the wallet balance; the only component permitted to credit/debit a wallet, always via a transaction record.
Transaction Manager	Orchestrates deposit/withdraw requests; contains one channel adapter per external partner (MTN, Orange, UBA); handles retries, timeouts, and reversals.
Database	Persistent store for users, wallets, linked accounts, transactions, and notifications (see Section 3.3).

3.3 Database Design

The data model is intentionally normalized around five entities: User, Wallet, Linked External Account (the user's MTN/Orange/UBA identifiers), Transaction, and Notification. Every wallet balance change must originate from a Transaction row — this is what makes the ledger auditable.

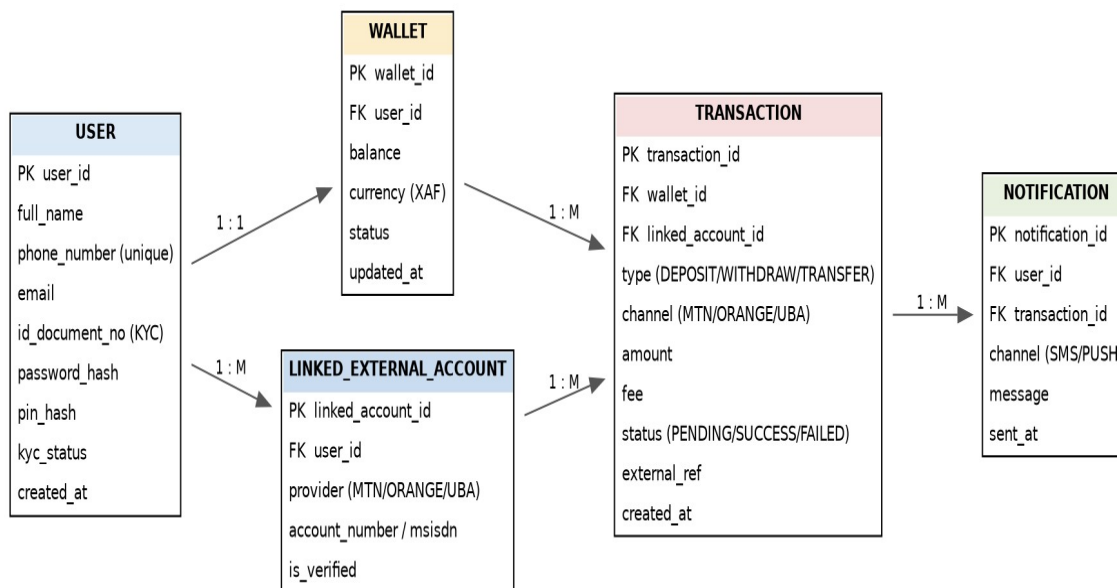


Figure 6: Entity-Relationship Diagram

3.3.1 Data dictionary — key tables

Table	Key Fields
User	user_id (PK), phone_number (unique), email, id_document_no, kyc_status
Wallet	wallet_id (PK), user_id (FK), balance, currency, status
Linked_External_Account	linked_account_id (PK), user_id (FK), provider [MTN/ORANGE/UBA], account_number/msisdn, is_verified
Transaction	transaction_id (PK), wallet_id (FK), linked_account_id (FK), type [DEPOSIT/WITHDRAW], channel, amount, fee, status [PENDING/SUCCESS/FAILED/REVERSED], external_ref
Notification	notification_id (PK), user_id (FK), transaction_id (FK), channel [SMS/PUSH], message, sent_at

3.4 Process Design — Transaction Sequence Flows

The two core money-movement operations — fund (deposit) and withdraw — follow the same generic pattern regardless of which of the three channels is used, which is the practical benefit of the Triangle model: the sequence below applies identically whether the channel is MTN Mobile Money, Orange Money, or UBA, with only the channel adapter changing underneath.

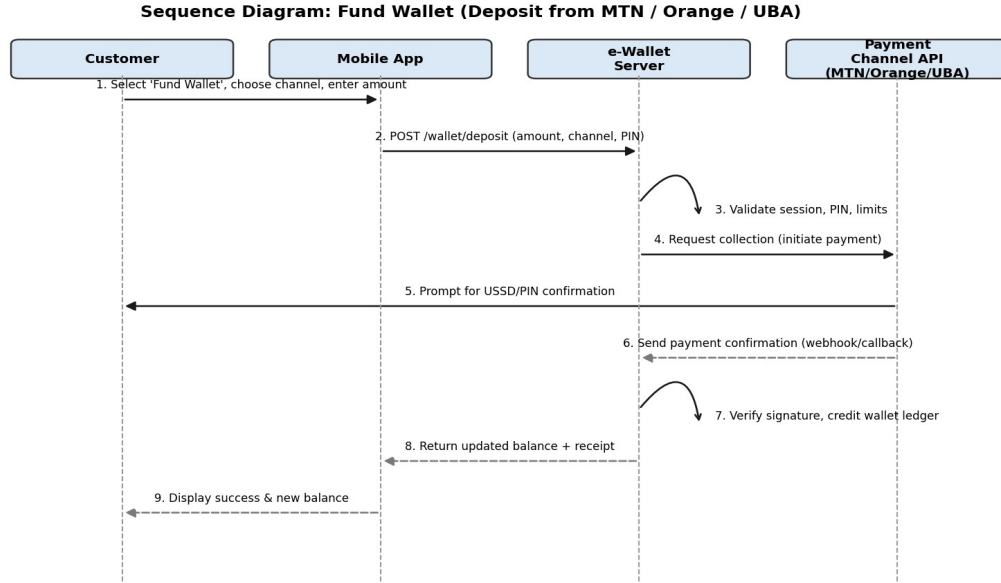


Figure 7: Sequence Diagram — Fund Wallet (Deposit)

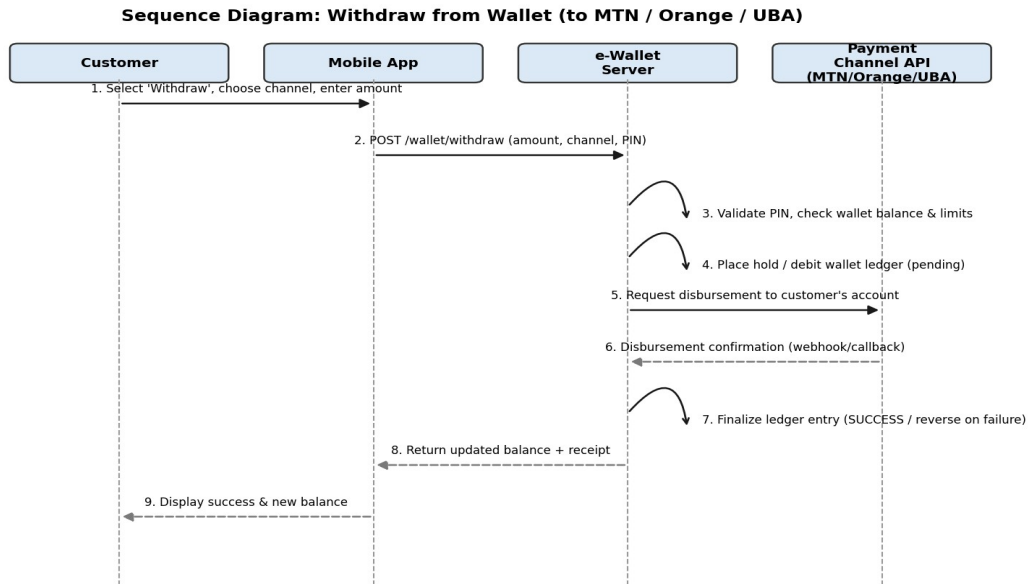


Figure 8: Sequence Diagram — Withdraw from Wallet

3.4.1 Transaction state machine

Every transaction moves through a small, well-defined set of states, which keeps reconciliation predictable:

- **PENDING** — created when the user submits a request; wallet funds are held (for withdrawals) but not yet finalised.
- **PROCESSING** — sent to the external channel; awaiting its callback/webhook.

- **SUCCESS** — channel confirmed completion; wallet ledger is finalised (credited for deposits, permanently debited for withdrawals).
- **FAILED** — channel rejected or timed out; any held funds are released back to the wallet automatically.
- **REVERSED** — a **SUCCESS** transaction that was later reversed (e.g., disputed or duplicate), always via a new offsetting transaction, never by editing history.

3.5 Interface / API Design

PayDay exposes a small, consistent internal REST API to its own mobile client, and consumes three external partner APIs through channel adapters.

3.5.1 Internal API (client-facing)

Endpoint	Method	Purpose
/auth/register	POST	Create a new user and KYC record; auto-creates the wallet.
/auth/login	POST	Authenticate and issue a session token (plus OTP step if enabled).
/wallet/balance	GET	Return the current wallet balance.
/wallet/deposit	POST	Initiate a deposit from a chosen channel (MTN / Orange / UBA).
/wallet/withdraw	POST	Initiate a withdrawal to a chosen channel.
/wallet/transactions	GET	Paginated transaction history.
/webhooks/mtn	POST	Callback endpoint for MTN MoMo payment/disbursement confirmations.
/webhooks/orange	POST	Callback endpoint for Orange Money confirmations.
/webhooks/uba	POST	Callback endpoint for UBA transaction confirmations.

3.5.2 External channel adapters

Channel	Typical integration	Deposit (collection) operation	Withdrawal (disbursement) operation
MTN Mobile Money	MTN MoMo API	Request to Pay — prompts	Transfer — disburses from

Channel	Typical integration	Deposit (collection) operation	Withdrawal (disbursement) operation
	(Collections & Disbursements, OAuth2 + API key/subscription key)	the customer to approve via MoMo USSD/PIN.	MTN's disbursement account to the customer's MoMo number.
Orange Money	Orange Money Web Payment / Orange Developer API	Web Payment initiation — customer confirms via Orange Money PIN prompt.	Merchant/aggregator payout API — credits the customer's Orange Money account.
UBA	UBA corporate/API banking channel (direct partnership or licensed aggregator)	Direct debit / account-to-account transfer request against the linked UBA account.	Bank transfer / credit instruction to the linked UBA account.

4. Security and Compliance Considerations

- KYC at registration: full name, phone number, and a valid national ID or passport number, in line with COBAC/BEAC electronic money and AML/CFT requirements.
- Every deposit and withdrawal requires a second factor: transaction PIN (stored as a hash, never in plaintext) and/or one-time password (OTP) sent by SMS.
- All client-server and server-to-partner traffic uses TLS 1.2 or higher; partner webhook payloads are verified using signatures/HMAC where the partner API supports it.
- Sensitive fields (ID numbers, linked account numbers) are encrypted at rest; passwords and PINs are hashed with a strong algorithm (bcrypt or argon2).
- Daily and monthly transaction limits per user, configurable by the administrator, to contain fraud exposure and align with e-money regulatory ceilings.
- Full audit logging: every state change on a transaction is logged with actor, timestamp, and reason, supporting dispute resolution and regulator reporting.
- Segregation of duties: the wallet's pooled funds should sit in a trust/float account at a licensed bank, reconciled daily against the sum of all wallet balances in the ledger.

5. Recommended Technology Stack

Layer	Suggested Technology	Rationale
Mobile client	Flutter (Android-first, given local device usage patterns)	Single codebase for Android/iOS; matches prior Flutter experience and Cameroon's largely Android user base.
Backend API	Node.js (Express/NestJS) or Python (Flask/FastAPI)	Mature ecosystems for REST APIs and webhook handling; large community support for MTN/Orange API integration examples.
Database	PostgreSQL or MySQL	ACID-compliant relational database, essential for a financial ledger.
Messaging/Queue	Redis or RabbitMQ	Buffers and retries deposit/withdraw requests to external channels; smooths out API rate limits and timeouts.
Notifications	SMS gateway (local aggregator) + Firebase Cloud Messaging	Reaches users even on basic phones (SMS) and provides rich push notifications on smartphones.
Hosting	Cloud VPS or PaaS with data residency consideration	Should account for Cameroonian/COBAC data handling expectations where applicable.

6. Implementation Roadmap

Phase	Milestone	Duration (indicative)
Phase 1	Core wallet, registration/KYC, database & security layer	3–4 weeks
Phase 2	MTN MoMo adapter (deposit & withdraw) + sandbox testing	2–3 weeks
Phase 3	Orange Money adapter (deposit & withdraw) + sandbox testing	2–3 weeks
Phase 4	UBA adapter (deposit & withdraw) + sandbox testing	2–4 weeks (subject to partner onboarding)
Phase 5	Reconciliation, admin back-office, notifications, and reporting	2 weeks
Phase 6	Pilot with a small user group, then production rollout	3–4 weeks

7. Conclusion

PayDay's Triangle model — one central wallet, three interoperable payment channels, two clear directions of money movement — provides a deliberately simple but extensible foundation for an e-banking application in the Cameroonian context. By isolating MTN Mobile Money, Orange Money, and UBA behind uniform channel adapters, and by treating every balance change as an auditable transaction, the system stays simple to build and reason about in version 1, while leaving a clear path to add wallet-to-wallet transfers, bill payments, and additional channels in future phases.

EVERYDAY IS A PAYDAY